

Quiz navigation

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	25
26	27	28	29	30
31	32	33	34	35
36	37	38	39	40
41	42	43	44	45
46	47	48	49	50
51	52	53	54	55
56	57	58	59	60
61	62	63	64	65
66	67	68	69	70
71	72	73	74	75
76	77	78	79	80
81	82	83	84	85
86	87	88	89	90
91	92	93	94	95
96	97	98	99	100
101	102	103	104	105
106	107	108	109	110
111	112	113	114	115
116	117	118	119	120

Show one page at a time

Finish review

Course Home

Course Info ▲

Syllabus

Participants list

Grade/Attendance ▲

Offline-Attendance



System Programming

6Week [16 March - 22 March]

SOC2040 SP QUIZ3 SPRING 2024

Started on	2024-03-28 19:02
State	Finished
Completed on	2024-03-29 23:58
Time taken	1 day 4 hours
Marks	525.00/1158.00
Grade	4.53 out of 10.00 (45%)

Question 1

Not answered

Marked out of 10.00

The instruction to perform add SIMD operation on Double-precision floating data is :

Select one:

- ☐ 1. addpd %xmm0, %xmm1
- ☐ 2. addsd %xmm0, %xmm1
- ☐ 3. addps %xmm0, %xmm1
- ☐ 4. addss %xmm0, %xmm1

The correct answer is: addpd %xmm0, %xmm1

Question 2

Correct

Mark 10.00 out of 10.00

Write x86-64 assembly language instructions to multiply x in %rax register by 1280 using only leaq and salq (shift left instructions)

Select one:

- ☒ 1. leaq(%rax, %rax, 4), %rax
salq \$8, %rax ✓
- ☐ 2. leaq(%rax, %rax, 1), %rax
salq \$9, %rax
- ☐ 3. leaq(%rax, %rax, 8), %rax
salq \$6, %rax
- ☐ 4. leaq(%rax, %rax, 2), %rax
salq \$7, %rax

The correct answer is: leaq(%rax, %rax, 4), %rax
salq \$8, %rax

Grades

Students Notifications ▾

Others ▾

Activities/Resources +

Question **3**

Correct

Mark 5.00 out of 5.00

Linux Command to convert C language program to corresponding assembly language program is

Select one:

- ☒ 1. gcc -Og -S mstore.c ✓
- ☐ 2. gcc -Og -c mstore.c
- ☐ 3. objdump -d mstore
- ☐ 4. gcc mstore.c -o mstore

The correct answer is: gcc -Og -S mstore.c

Question **4**

Correct

Mark 15.00 out of 15.00

ISA stands for

Answer: INSTRUCTION SET ARCHITECTURE ✓

The correct answer is: Instruction Set Architecture

Question **5**

Correct

Mark 10.00 out of 10.00

..... instruction pops address from the stack top pointed to by %rsp into %rip

Answer: ret ✓

The correct answer is: ret

Question **6**

Correct

Mark 10.00 out of 10.00

Full form of RISC is

Answer: Reduced Instruction Set Computer ✓

The correct answer is: Reduced Instruction Set Computer

Question **7**

Correct

Mark 5.00 out of 5.00

To invoke sys_write kernel function, the function id to be loaded into %rax register is

Select one:

- ☐ 1. 0
- ☒ 2. 1 ✓
- ☐ 3. 60
- ☐ 4. 2

The correct answer is: 1

Question **8**
Correct
Mark 15.00 out of 15.00

FULL FORM OF AVX

Answer: Advanced Vector Extensions ✓

The correct answer is: ADVANCED VECTOR EXTENSIONS

Question **9**
Correct
Mark 10.00 out of 10.00

The arguments to be passed to the C library `printf(format, arg)` function before calling the `printf` function in assembly language program to display an arg integer value.

Select one:

- ☐ 1. `%rdi = $format, %rdx= arg, %rbx=0`
- ☐ 2. `%rdi = $format, %rsi= arg, %rbx=0`
- ☐ 3. `%rsi = $format, %rdi= arg, %rax=0`
- ☒ 4. `%rdi = $format, %rsi= arg, %rax=0` ✓

The correct answer is: `%rdi = $format, %rsi= arg, %rax=0`

Question **10**
Correct
Mark 5.00 out of 5.00

IP stands for

Answer: Instruction Pointer ✓

The correct answer is: Instruction Pointer

Question **11**
Correct
Mark 5.00 out of 5.00

The instruction `testq %rbx, %rax` performs the following operation:

Select one:

- ☐ 1. `%rax - %rbx`
- ☐ 2. `%rax ^ %rbx`
- ☒ 3. `%rax & %rbx` ✓
- ☐ 4. `%rax | %rbx`

The correct answer is: `%rax & %rbx`

Question 12

Correct

Mark 5.00 out of 5.00

Assembly language instruction to move 32-bits of data from the memory location pointed to by %rsi register to a register is

Select one:

- ☐ 1. movl (%rsi), ebx
- ☒ 2. movl (%rsi), %ebx ✓
- ☐ 3. movw (%rsi), %bx
- ☐ 4. movq (%rsi), %rbx

The correct answer is: movl (%rsi), %ebx

Question 13

Correct

Mark 5.00 out of 5.00

To invoke sys_read kernel function, the function id to be loaded into %rax register is

Select one:

- ☐ 1. 1
- ☒ 2. 0 ✓
- ☐ 3. 2
- ☐ 4. 60

The correct answer is: 0

Question 14

Correct

Mark 15.00 out of 15.00

Indicate the value computed in register %rax by the following x86-64 assembly language recursive function if the register %rdi contains 0xFEEDBAC5 initially.

rec_fun:

```
movq $0, %rax
testq %rdi, %rdi
je DOWN
pushq %rbx
movq %rdi, %rbx
andq $1, %rbx
shrq %rdi
call rec_fun
addq %rbx, %rax
popq %rbx
```

DOWN: ret

Select one:

- ☐ 1. 32
- ☒ 2. 22 ✓
- ☐ 3. 42
- ☐ 4. 12

The correct answer is: 22

Question **15**

Correct

Mark 5.00 out of 5.00

A quad word contains

Select one:

- ☐ 1. 8 bits
- ☒ 2. 64 bits ✓
- ☐ 3. 32 bits
- ☐ 4. 16 bits

The correct answer is: 64 bits

Question **16**

Correct

Mark 5.00 out of 5.00

Assembly language instruction to move the immediate data 0xdadcab78 to register %rcx is

Select one:

- ☐ 1. movl \$0xdadcab78, %rcx
- ☐ 2. movq %rcx, \$0xdadcab78
- ☒ 3. movq \$0xdadcab78, %rcx ✓
- ☐ 4. movq \$0xdadcab78, rcx

The correct answer is: movq \$0xdadcab78, %rcx

Question **17**

Correct

Mark 5.00 out of 5.00

If %rbx = 9000 and %rsi = 50 then determine the Effective Address computed in the following instruction:

movq -98(%rbx, %rsi, 4), %rax

Select one:

- ☐ 1. 0x9298
- ☒ 2. 9102 ✓
- ☐ 3. 0x9102
- ☐ 4. 9298

The correct answer is: 9102

Question **18**

Correct

Mark 5.00 out of 5.00

The ZMM registers in AVX technology are all bits wide.

Select one:

- ☐ 1. 64
- ☒ 2. 512 ✓
- ☐ 3. 128
- ☐ 4. 256

The correct answer is: 512

Question **19**
Correct
Mark 10.00 out of 10.00

The condition checked in `jle` (jump on less than or equal to) conditional jump instruction is

Select one:

- ☐ 1. $(SF \wedge OF)$
- ☐ 2. $\sim(SF \wedge OF) \& \sim ZF$
- ☒ 3. $(SF \wedge OF) | ZF$ ✓
- ☐ 4. $\sim(SF \wedge OF)$

The correct answer is: $(SF \wedge OF) | ZF$

Question **20**
Correct
Mark 5.00 out of 5.00

Given `%rdx = 0xa000`, `%rsi=16`, determine the value stored in register `%rax` after the execution of the following instruction :

`leaq(%rdx, %rsi, 8), %rax`

Select one:

- ☒ 1. `0xa080` ✓
- ☐ 2. `0xa128`
- ☐ 3. `0xa040`
- ☐ 4. `0xa064`

The correct answer is: `0xa080`

Question **21**
Correct
Mark 10.00 out of 10.00

The condition checked in `jl` (jump on less than) conditional jump instruction is

Select one:

- ☐ 1. $(SF \wedge OF) | ZF$
- ☐ 2. $\sim(SF \wedge OF)$
- ☐ 3. $\sim(SF \wedge OF) \& \sim ZF$
- ☒ 4. $(SF \wedge OF)$ ✓

The correct answer is: $(SF \wedge OF)$

Question **22**
Not answered
Marked out of 10.00

FULL FORM OF REGISTER SI

Answer: ✗

The correct answer is: SOURCE INDEX

Question **23**

Correct

Mark 5.00 out of 5.00

..... is the first 64-bit intel x86 processor referred to as x86-64 introduced in 2004.

Select one:

- ☐ 1. Opetron
- ☐ 2. Itanium
- ☐ 3. Pentium
- ☒ 4. Pentium 4E ✓

The correct answer is: Pentium 4E

Question **24**

Incorrect

Mark 0.00 out of 15.00

FULL FORM OF SIMD

Answer: STREAMING SIMD EXTENSIONS



The correct answer is: SINGLE INSTRUCTION MULTIPLE DATA

Question **25**

Correct

Mark 5.00 out of 5.00

..... processor from AMD was a tough competitor to intel Pentium 4.

Select one:

- ☐ 1. Pentium 4E
- ☐ 2. Itanium
- ☐ 3. Core 2
- ☒ 4. Opetron ✓

The correct answer is: Opetron

Question **26**

Not answered

Marked out of 10.00

The instruction to perform add SIMD operation on Double-precision floating data is :

Select one:

- ☐ 1. addpd %xmm0, %xmm1
- ☐ 2. addss %xmm0, %xmm1
- ☐ 3. addps %xmm0, %xmm1
- ☐ 4. addsd %xmm0, %xmm1

The correct answer is: addpd %xmm0, %xmm1

Question **27**

Correct

Mark 5.00 out of 5.00

For 32-bit signed division on a 32-bit intel processor, instruction is used to sign-extend 32-bit dividend in %eax register into %edx to make the dividend 64-bit in %edx:%eax

Select one:

- ☐ 1. cqo
- ☐ 2. cwtq
- ☐ 3. cqto
- ☒ 4. cltd ✓

The correct answer is: cltd

Question **28**

Correct

Mark 10.00 out of 10.00

FULL FORM OF MMX

Select one:

- ☐ 1. Memory Management Extensions
- ☐ 2. Modern Multiprocessor Xeon
- ☒ 3. Multi-Media Extensions ✓
- ☐ 4. Main Memory Extensions

The correct answer is: Multi-Media Extensions

Question **29**

Correct

Mark 20.00 out of 20.00

Write x86-64 assembly language instructions to multiply x in %rax register by 3072 using only leaq and salq (shift left instructions)

Select one:

- ☐ 1. leaq(%rax, %rax, 8), %rax
salq \$7, %rax
- ☒ 2. leaq(%rax, %rax, 2), %rax
salq \$10, %rax ✓
- ☐ 3. leaq(%rax, %rax, 4), %rax
salq \$9, %rax
- ☐ 4. leaq(%rax, %rax, 1), %rax
salq \$12, %rax

The correct answer is: leaq(%rax, %rax, 2), %rax
salq \$10, %rax

Question **30**
Not answered
Marked out of 10.00

The instruction to perform add scalar operation on Double-precision floating data is :

Select one:

- ☐ 1. addpd %xmm0, %xmm1
- ☐ 2. addss %xmm0, %xmm1
- ☐ 3. addsd %xmm0, %xmm1
- ☐ 4. addps %xmm0, %xmm1

The correct answer is: addsd %xmm0, %xmm1

Question **31**
Correct
Mark 20.00 out of 20.00

Write the function performed by the following recursive function `rec_fun`.

`rec_fun`:

```
movq $0, %rax
testq %rdi, %rdi
je DOWN
pushq %rbx
movq %rdi, %rbx
andq $1, %rbx
shrq %rdi
call rec_fun
addq %rbx, %rax
popq %rbx
```

DOWN: `ret`

Select one:

- ☒ 1. Counting the number of 1's in %rdi ✓
- ☐ 2. Computing the factorial of number in %rdi
- ☐ 3. Dividing the contents of %rdi by 64
- ☐ 4. Computing addition of numbers in %rax & %rbx

The correct answer is: Counting the number of 1's in %rdi

Question **32**
Correct
Mark 5.00 out of 5.00

A long word contains

Select one:

- ☐ 1. 16 bits
- ☒ 2. 32 bits ✓
- ☐ 3. 64 bits
- ☐ 4. 8 bits

The correct answer is: 32 bits

Question **33**

Correct

Mark 10.00 out of 10.00

For the following 64-bit signed division operation on a 64-bit intel processor, the 64-bit quotient is stored in register and the 64-bit remainder is stored in

`idivq %rbx`

Select one:

- ☐ 1. %rbx(remainder), %rax(quotient)
- ☐ 2. %rax(remainder), %rbx(quotient)
- ☐ 3. %rax(remainder), %rdx(quotient)
- ☒ 4. %rdx(remainder), %rax(quotient) ✓

The correct answer is: %rdx(remainder), %rax(quotient)

Question **34**

Correct

Mark 10.00 out of 10.00

Indicate the function performed by the following x86-64 assembly language program segment:

`xorq %rbx, %rax`

`xorq %rax, %rbx`

`xorq %rbx, %rax`

Select one:

- ☐ 1. Exclusive or of the contents of the register %rax to %rbx
- ☐ 2. Subtraction of two numbers in registers %rax and %rbx
- ☐ 3. Addition of two numbers in registers %rax and %rbx
- ☒ 4. Exchange of the contents of registers %rax with %rbx without using any temporary register or memory location or stack ✓

The correct answer is: Exchange of the contents of registers %rax with %rbx without using any temporary register or memory location or stack

Question **35**

Correct

Mark 5.00 out of 5.00

A word contains

Select one:

- ☐ 1. 32 bits
- ☐ 2. 8 bits
- ☐ 3. 64 bits
- ☒ 4. 16 bits ✓

The correct answer is: 16 bits

Question 36

Correct

Mark 10.00 out of 10.00

Indicate the value computed in register %rax by the following x86-64 assembly language recursive function if n=8 initially.

```

movq    n, %rbx
movq    %rbx,%rax
call    recp
.....
.....

```

recp:

```

cmpq    $1, %rbx
je      OUT
decq    %rbx
pushq   %rbx
call    recp
popq    %rbx
mulq    %rbx

```

OUT: ret

Select one:

- ☐ 1. 362880
- ☒ 2. 40320 ✓
- ☐ 3. 720
- ☐ 4. 5040

The correct answer is: 40320

Question 37

Correct

Mark 10.00 out of 10.00

Given %rdx = 0xFDEA0000, %rsi=20, determine the value stored in register %rax after the execution of the following instruction :

```
leaq (%rdx, %rsi, 8), %rax
```

Select one:

- ☐ 1. 0x 0xFDEA0160
- ☐ 2. 0x 0xFDEA0020
- ☒ 3. 0x 0xFDEA00A0 ✓
- ☐ 4. 0x 0xFDEA0008

The correct answer is: 0x 0xFDEA00A0

Question 38

Correct

Mark 10.00 out of 10.00

The condition checked in jg (jump on greater than) conditional jump instruction is

Select one:

- ☐ 1. (SF^OF)
- ☒ 2. ~(SF^OF)&~ZF ✓
- ☐ 3. (SF^OF)|ZF
- ☐ 4. ~(SF^OF)

The correct answer is: ~(SF^OF)&~ZF

Question **39**

Correct

Mark 10.00 out of 10.00

Given %rdx = 0xFDEA0000, %rsi=40, determine the value stored in register %rax after the execution of the following instruction :

`leaq 0x200(%rdx, %rsi, 8), %rax`

Select one:

- ☐ 1. 0xFDEA0240
- ☐ 2. 0xFDEA0520
- ☐ 3. 0xFDEA0248
- ☒ 4. 0xFDEA0340 ✓

The correct answer is: 0xFDEA0340

Question **40**

Correct

Mark 10.00 out of 10.00

Given %rdx = 0xA0000, %rsi=20, determine the value stored in register %rax after the execution of the following instruction :

`leaq 0xFDEA0000(%rdx, %rsi, 8), %rax`

Select one:

- ☐ 1. 0xFDEA0014
- ☒ 2. 0xFDF400A0 ✓
- ☐ 3. 0xFDEA0020
- ☐ 4. 0xFDEA0160

The correct answer is: 0xFDF400A0

Question **41**

Correct

Mark 5.00 out of 5.00

The XMM registers in SSE technology are all bits wide

Select one:

- ☐ 1. 64
- ☐ 2. 256
- ☐ 3. 512
- ☒ 4. 128 ✓

The correct answer is: 128

Question 42

Correct

Mark 10.00 out of 10.00

Given the following structure stored in the computer memory, determine the alignment requirement for the structure, i.e., the address of the structure and length of the structure must be a multiple of K. What is K in this structure?

```
struct S1 {  
    char c;  
    char d;  
    double v;  
    char e;  
    double w;  
    char f;  
} *p;
```

Select one:

- ☐ 1. 16
- ☐ 2. 4
- ☒ 3. 8 ✓
- ☐ 4. 2

The correct answer is: 8

Question 43

Correct

Mark 20.00 out of 20.00

Indicate the value computed in register %rax by the following x86-64 assembly language recursive function if n=8 initially.

```
movq    n, %rbx  
    movq    %rbx,%rax  
    call    recp  
    .....  
    .....  
recp:  
    cmpq    $1, %rbx  
    je      OUT  
    decq    %rbx  
    pushq   %rbx  
    call    recp  
    popq    %rbx  
    mulq    %rbx  
OUT:    ret
```

Select one:

- ☐ 1. 720
- ☐ 2. 362880
- ☐ 3. 5040
- ☒ 4. 40320 ✓

The correct answer is: 40320

Question 44

Correct

Mark 5.00 out of 5.00

..... is the first multi-core intel x86 processor introduced in 2006.

Select one:

- ☐ 1. Core i7
- ☒ 2. Core 2 ✓
- ☐ 3. Core i9
- ☐ 4. Core i5

The correct answer is: Core 2

Question 45

Correct

Mark 20.00 out of 20.00

Given a matrix $a[R][C]$ containing elements of type double stored in the computer main memory in Row-major order where R is the total number of rows and C is the total number of Columns in the matrix.

Choose the correct program segment containing x86-64 assembly language instructions to compute the address of the location containing the matrix element $a[i][j]$ where i is the row index & j is the column index.

Given Register Mapping: $\%rdi \leftarrow a$, $\%rax \leftarrow C$, $\%rcx \leftarrow R$, $\%rsi \leftarrow j$, $\%rbx \leftarrow i$

Select one:

- ☐ 1. `leaq (%rax, 8), %rax`
`leaq (%rax, 8), %rax`
`mulq %rbx`
`leaq (%rbx, %rsi), %rax`
- ☒ 2. `leaq (%rdi, %rsi, 8), %rsi`
`mulq %rbx`
`leaq (%rsi, %rax, 8), %rax` ✓
- ☐ 3. `leaq (%rdi, %rsi, 8), %rsi`
`mulq %rbx`
`leaq(%rax, %rsi, 8), %rax`
- ☐ 4. `leaq (%rax, 8), %rax`
`leaq(%rdi, %rsi, 8) %rsi`
`mulq %rax`
`leaq (%rax, %rsi), %rax`

The correct answer is: `leaq (%rdi, %rsi, 8), %rsi`
`mulq %rbx`
`leaq (%rsi, %rax, 8), %rax`

Question 46

Correct

Mark 5.00 out of 5.00

The arguments to be passed to the syscall `sys_exit` function are

Select one:

- ☐ 1. $\%rax=0$, $\%rdi=60$, $\%rsi$ = Buffer starting address, $\%rdx$ =Buffer size
- ☐ 2. $\%rax=60$, $\%rdi=0$, $\%rdx$ = Buffer starting address, $\%rsi$ =Buffer size
- ☐ 3. $\%rax=0$, $\%rdi=60$
- ☒ 4. $\%rax=60$, $\%rdi=0$ ✓

The correct answer is: $\%rax=60$, $\%rdi=0$

Question **47**

Not answered

Marked out of 5.00

A statement in C provides a multi-way branching capability based on the value of an integer index.

Answer:



The correct answer is: switch

Question **48**

Not answered

Marked out of 20.00

Indicate the function performed by the following x86-64 assembly language program segment :

```
addq %rbx, %rax
subq %rax, %rbx
negq %rbx
subqq %rbx, %rax
```

Select one:

- ☐ 1. Exchange of the contents of registers %rax with %rbx without using any temporary register or memory location or stack
- ☐ 2. Addition of two numbers in registers %rax and %rbx
- ☐ 3. Exclusive or of the contents of the register %rax to %rbx
- ☐ 4. Subtraction of two numbers in registers %rax and %rbx

The correct answer is: Exchange of the contents of registers %rax with %rbx without using any temporary register or memory location or stack

Question **49**

Incorrect

Mark 0.00 out of 10.00

The callq ffact instruction performs the following operations :

Select one:

- ☐ 1. jmp ffact
pushq %rip
- ☒ 2. popq %rip
jmp ffact
- ☐ 3. jmp ffact
popq %rip
- ☐ 4. pushq %rip
jmp ffact

The correct answer is: pushq %rip
jmp ffact

Question 50

Correct

Mark 10.00 out of 10.00

Write the function performed by the following recursive procedure `recp`.

```
movq    n, %rbx
movq    %rbx,%rax
call    recp
.....
.....
recp:
    cmpq    $1, %rbx
    je      OUT
    decq    %rbx
    pushq   %rbx
    call    recp
    popq    %rbx
    mulq    %rbx
OUT:    ret
```

Select one:

- ☐ 1. Computing addition of numbers in %rax & %rbx
- ☒ 2. Computing the factorial of number n in %rbx ✓
- ☐ 3. Multiplying the contents of %rbx by %rax
- ☐ 4. Counting the number of 1's in n at %rbx

The correct answer is: Computing the factorial of number n in %rbx

Question 51

Correct

Mark 10.00 out of 10.00

Write x86-64 assembly language instructions to multiply x in %rax register by 144 using only `leaq` and `shlq` (shift left instructions)

Select one:

- ☒ 1. `leaq(%rax, %rax, 8), %rax`
`shlq $4, %rax` ✓
- ☐ 2. `leaq(%rax, %rax, 1), %rax`
`shlq $7, %rax`
- ☐ 3. `leaq(%rax, %rax, 2), %rax`
`shlq $6, %rax`
- ☐ 4. `leaq(%rax, %rax, 4), %rax`
`shlq $5, %rax`

The correct answer is: `leaq(%rax, %rax, 8), %rax`
`shlq $4, %rax`

Question **52**

Correct

Mark 10.00 out of 10.00

The arguments to be passed to the C library `scanf(format, &arg)` function before calling the `scanf` function in assembly language program to read an arg integer value from the keyboard.

Select one:

- ☒ 1. `%rdi = $format, %rsi= &arg, %rax=0` ✓
- ☐ 2. `%rdi = $format, %rsi= &arg, %rbx=0`
- ☐ 3. `%rsi = $format, %rdi= &arg, %rax=0`
- ☐ 4. `%rdi = $format, %rsi= arg, %rax=0`

The correct answer is: `%rdi = $format, %rsi= &arg, %rax=0`

Question **53**

Correct

Mark 5.00 out of 5.00

For 64-bit signed division on a 64-bit intel processor, instruction is used to sign-extend 64-bit dividend in `%rax` register into `%rdx` to make the dividend 128-bit in `%rdx:%rax`

Select one:

- ☐ 1. `cqtd`
- ☐ 2. `cwtq`
- ☒ 3. `cqo` ✓
- ☐ 4. `cltd`

The correct answer is: `cqo`

Question **54**

Correct

Mark 10.00 out of 10.00

DI stands for

Answer: ✓

The correct answer is: Destination Index

Question **55**
Correct
Mark 10.00 out of 10.00

Given the following structure stored in the computer memory starting at address 0x2000, determine the length of the structure after satisfying the alignment requirement for the elements in the structure.

```
struct S1 {  
    char c;  
    char d;  
    double v;  
    char e;  
    double w;  
    char f;  
} *p;
```

Select one:

- ☐ 1. 28 bytes
- ☐ 2. 32 bytes
- ☒ 3. 40 bytes ✓
- ☐ 4. 20 bytes

The correct answer is: 40 bytes

Question **56**
Correct
Mark 5.00 out of 5.00

The intel assembly language instruction to invoke system call is

Answer: ✓

The correct answer is: syscall

Question **57**
Correct
Mark 10.00 out of 10.00

What is the data stored in %rbx register after the execution of following instruction:

```
movq $0x8bcdef8960, %rbx
```

Select one:

- ☐ 1. 0x1000008bcdef8960
- ☐ 2. 0x0000008bcdef8960
- ☐ 3. 0x1111118bcdef8960
- ☒ 4. 0xffffffff8bcdef8960 ✓

The correct answer is: 0xffffffff8bcdef8960

Question **58**
Correct
Mark 10.00 out of 10.00

Write x86-64 assembly language instructions to multiply x in %rax register by 1024 using only leaq and shlq (shift left instructions)

Select one:

- ☐ 1. leaq(%rax, %rax, 8), %rax
shlq \$5, %rax
- ☒ 2. leaq(%rax, %rax, 1), %rax
shlq \$9, %rax ✓
- ☐ 3. leaq(%rax, %rax, 2), %rax
shlq \$8, %rax
- ☐ 4. leaq(%rax, %rax, 4), %rax
shlq \$6, %rax

The correct answer is: leaq(%rax, %rax, 1), %rax
shlq \$9, %rax

Question **59**
Correct
Mark 5.00 out of 5.00

Indicate the function performed by the following x86-64 assembly language program segment.

```
movq  (%rdi), %rax
movq  (%rsi), %rdx
movq  %rdx, (%rdi)
movq  %rax, (%rsi)
```

Select one:

- ☐ 1. Copying the contents of the register %rax to %rdx
- ☐ 2. Copying the contents of the register %rdx to %rax
- ☐ 3. Exchange of the contents of registers %rax and %rdx using memory locations
- ☒ 4. Exchange of the contents of memory locations pointed to by registers %rdx and %rsi using temporary registers ✓

The correct answer is: Exchange of the contents of memory locations pointed to by registers %rdx and %rsi using temporary registers

Question **60**
Correct
Mark 10.00 out of 10.00

FULL FORM OF REGISTER BP

Answer: ✓

The correct answer is: BASE POINTER

Question **61**

Correct

Mark 5.00 out of 5.00

Identify the condition codes or flags in intel x86-64 processor which store status information about most recent arithmetic and logical operations.

Select one:

- ☐ 1. SF, IF, ZF, CF
- ☐ 2. TF, IF, CF, OF
- ☒ 3. SF, CF, OF, ZF ✓
- ☐ 4. OF, TF, CF, ZF

The correct answer is: SF, CF, OF, ZF

Question **62**

Incorrect

Mark 0.00 out of 20.00

Reorder and Provide the correct Steps in Designing Recursive Function:

1. Identify & compute the next argument to be passed to the recursive function
2. Identify & push intermediate result onto the stack
3. Call the recursive function with the next argument
4. Identify and write the tasks to be performed in the recursive function body
5. Identify and write the tasks to be performed when the recursive function returns
6. Identify & write the termination condition for the recursive function

Select one:

- ☒ 1. 1, 2, 4, 3, 6, 5 ✗
- ☐ 2. 1, 2, 3, 4, 5, 6
- ☐ 3. 6, 4, 1, 2, 5, 3
- ☐ 4. 6, 4, 2, 1, 3, 5

The correct answer is: 6, 4, 2, 1, 3, 5

Question **63**

Incorrect

Mark 0.00 out of 10.00

The instruction to perform add scalar operation on single-precision floating data is :

Select one:

- ☒ 1. addsd %xmm0, %xmm1 ✗
- ☐ 2. addps %xmm0, %xmm1
- ☐ 3. addpd %xmm0, %xmm1
- ☐ 4. addss %xmm0, %xmm1

The correct answer is: addss %xmm0, %xmm1

Question **64**

Correct

Mark 5.00 out of 5.00

The syscall invokes the interrupt instruction.

Select one:

- ☐ 1. int 0x128
- ☐ 2. int 80
- ☐ 3. int 32
- ☒ 4. int 0x80 ✓

The correct answer is: int 0x80

Question **65**

Correct

Mark 10.00 out of 10.00

Write a single assembly language instruction which performs the same operation as performed by the following program segment :

```
movq %rax, %rcx
movq %rbx, %rdx
notq %rcx
notq %rdx
andq %rdx, %rax
andq %rcx, %rbx
orq %rbx, %rax
```

Select one:

- ☒ 1. xorq %rbx, %rax ✓
- ☐ 2. subq %rbx, %rax
- ☐ 3. xorq %rcx, %rdx
- ☐ 4. addq %rbx, %rax

The correct answer is: xorq %rbx, %rax

Question **66**

Correct

Mark 15.00 out of 15.00

Full form of CISC is

Answer: ✓

The correct answer is: Complex Instruction Set Computer

Question **67**
Correct
Mark 15.00 out of 15.00

Assume that `%rsp = 0x120` and `%rip=0x400544` and you are given the following program segment of the main function:

address

0x400544: `callq 400550`

0x400549: `movq %rax, (%rbx)`

Write the contents of the registers `%rsp` and `%rip` after the execution of the instruction `callq` at address 0x400544

Select one:

- ☐ 1. `%rsp=0x128, %rip=400550`
- ☒ 2. `%rsp=0x118, %rip=400550` ✓
- ☐ 3. `%rsp=0x118, %rip=400549`
- ☐ 4. `%rsp=0x128, %rip=400549`

The correct answer is: `%rsp=0x118, %rip=400550`

Question **68**
Correct
Mark 10.00 out of 10.00

For the following 64-bit signed multiplication operation on a 64-bit intel processor, the 128-bit product result is stored in register pairs.

`imulq %rbx`

Select one:

- ☐ 1. `%rax(higher 64-bit):%rbx(lower 64-bit)`
- ☒ 2. `%rdx(higher 64-bit):%rax(lower 64-bit)` ✓
- ☐ 3. `%rax(higher 64-bit):%rdx(lower 64-bit)`
- ☐ 4. `%rbx(higher 64-bit):%rax(lower 64-bit)`

The correct answer is: `%rdx(higher 64-bit):%rax(lower 64-bit)`

Question **69**
Not answered
Marked out of 5.00

..... is a set of memory locations set aside in main memory to be accessed in LIFO fashion.

Answer:



The correct answer is: stack

Question **70**

Not answered

Marked out of
10.00

Given the following assembly language program segment, write the value in register %rax after the execution of this program segment.

```
movq    $0x01289AB76EF34567, %rax
movb    $0xBB, %dl
movsbq  %dl, %rax
```

Select one:

- ☐ 1. %rax = 0x01289AB76EF3BBBB
- ☐ 2. %rax = 0xFFFFFFFFFFFFBB
- ☐ 3. %rax = 0x01289AB76EF345BB
- ☐ 4. %rax = 0x00000000000000BB

The correct answer is: %rax = 0xFFFFFFFFFFFFBB

Question **71**

Not answered

Marked out of
20.00

Given %eax = 0x6DEAFFFF, %ebx = 0x9DEA0000, and flag CF=ZF=OF=SF=0 initially, determine the value stored in registers %eax, %ebx and flags after the execution of the following instruction :

```
cmpl %ebx, %eax
```

Select one:

- ☐ 1. %eax = 0x6DEAFFFF, %ebx = 0x9DEA0000, flag CF=0, ZF=0, OF=1, SF=1
- ☐ 2. %eax = 0xD000FFFF, %ebx = 0x9DEA0000, flag CF=1, ZF=0, OF=1, SF=1
- ☐ 3. %eax = 0x6BXD4FFFF, %ebx = 0x9DEA0000, flag CF=0, ZF=0, OF=0, SF=0
- ☐ 4. %eax = 0x6DEA0000, %ebx = 0x9DEAFFFF, flag CF=1, ZF=0, OF=1, SF=1

The correct answer is: %eax = 0x6DEAFFFF, %ebx = 0x9DEA0000, flag CF=0, ZF=0, OF=1, SF=1

Question **72**

Not answered

Marked out of
15.00

FULL FORM OF SSE

Answer:



The correct answer is: STREAMING SIMD EXTENSIONS

Question **73**

Not answered

Marked out of
20.00

Given initially %rax=100 & %rbx=200, write the values in the registers %rax & %rbx after the execution of the following x86-64 assembly language program segment

```
addq %rbx, %rax
subq %rax, %rbx
negq %rbx
subqq %rbx, %rax
```

Select one:

- ☐ 1. %rax= 200, %rbx = 100
- ☐ 2. %rax= 100, %rbx = 100
- ☐ 3. %rax= 200, %rbx = 300
- ☐ 4. %rax= 300, %rbx = 200

The correct answer is: %rax= 200, %rbx = 100

Question **74**

Not answered

Marked out of
10.00

The arguments to be passed to the syscall Display unit sys_write function are

Select one:

- ☐ 1. %rax=0, %rdi=1, %rdx = Buffer starting address, %rsi=Buffer size
- ☐ 2. %rax=1, %rdi=1, %rdx = Buffer starting address, %rsi=Buffer size
- ☐ 3. %rax=0, %rdi=0, %rdx = Buffer starting address, %rsi=Buffer size
- ☐ 4. %rax=1, %rdi=1, %rsi = Buffer starting address, %rdx=Buffer size

The correct answer is: %rax=1, %rdi=1, %rsi = Buffer starting address, %rdx=Buffer size

Question **75**

Not answered

Marked out of
10.00

If %rbx = 0x9000 and %rsi =0x50 then determine the Effective Address computed in the following instruction:

```
movq -98(%rbx, %rsi, 4), %rax
```

Select one:

- ☐ 1. 0x9702
- ☐ 2. 0x90de
- ☐ 3. 9298
- ☐ 4. 0x9222

The correct answer is: 0x90de

Question **76**

Not answered

Marked out of
10.00

The arguments to be passed to the syscall keyboard sys_read function are

Select one:

- ☐ 1. %rax=0, %rdi=0, %rdx = Buffer starting address, %rsi=Buffer size
- ☐ 2. %rax=0, %rdi=1, %rdx = Buffer starting address, %rsi=Buffer size
- ☐ 3. %rax=0, %rdi=0, %rsi = Buffer starting address, %rdx=Buffer size
- ☐ 4. %rax=1, %rdi=0, %rsi = Buffer starting address, %rdx=Buffer size

The correct answer is: %rax=0, %rdi=0, %rsi = Buffer starting address, %rdx=Buffer size

Question **77**

Not answered

Marked out of
6.00

Given the following assembly language program segment, write the value in register %rax after the execution of this program segment.

```
movq $0x01289AB76EF34567, %rax
```

```
movq    $-1, %rax
```

Select one:

- ☐ 1. %rax = 0xFFFFFFFFFFFFFFFF
- ☐ 2. %rax = 0x00000000FFFFFFFF
- ☐ 3. %rax = 0x01289AB76EF345FF
- ☐ 4. %rax = 0x01289AB76EF3FFFF

The correct answer is: %rax = 0xFFFFFFFFFFFFFFFF

Question **78**

Not answered

Marked out of
10.00

Given %rsi= 2000 which is the starting address of the memory area, Then determine how data is stored in the memory of a Intel machine after the execution of the following instructions:

```
movq $0xcabdadfeed123456, %rax
```

```
movq %rax, (%rsi)
```

Select one:

- ☐ 1. 2000:65, 2001:43 , 2002:21, 2003:de, 2004:ef, 2005:da, 2006:db, 2007:ac
- ☐ 2. 2000:ca, 2001:bd, 2002:ad, 2003:fe, 2004:ed, 2005:12, 2006:34, 2007:56
- ☐ 3. 2000:56, 2001:34, 2002:12, 2003:ed, 2004:fe, 2005:ad, 2006:bd, 2007:ca
- ☐ 4. 2000:ac, 2001:db, 2002:da, 2003:ef, 2004:de, 2005:21, 2006:43, 2007:65

The correct answer is: 2000:56, 2001:34, 2002:12, 2003:ed, 2004:fe, 2005:ad, 2006:bd, 2007:ca

Question **79**

Not answered

Marked out of
6.00

Given the following assembly language program segment, write the value in register %rax after the execution of this program segment.

```
movq $0x01289AB76EF34567, %rax
```

```
movw $-1, %ax
```

Select one:

- ☐ 1. %rax = 0x00000000FFFFFFFF
- ☐ 2. %rax = 0xFFFFFFFFFFFFFFFF
- ☐ 3. %rax = 0x01289AB76EF3FFFF
- ☐ 4. %rax = 0x01289AB76EF345FF

The correct answer is: %rax = 0x01289AB76EF3FFFF

Question **80**

Not answered

Marked out of
20.00

Given %eax = 0x6DEAFFFF, %ebx = 0x9DEA0000, and flag CF=ZF=OF=SF=0 initially, determine the value stored in registers %eax, %ebx and flags after the execution of the following instruction :

```
subl %ebx, %eax
```

Select one:

- ☐ 1. %eax = 0xD000FFFF, %ebx = 0x9DEA0000, flag CF=0, ZF=0, OF=1, SF=1
- ☐ 2. %eax = 0x6DEAFFFF, %ebx = 0x9DEA0000, flag CF=1, ZF=0, OF=1, SF=1
- ☐ 3. %eax = 0x6BXD4FFFF, %ebx = 0x9DEA0000, flag CF=0, ZF=0, OF=0, SF=0
- ☐ 4. %eax = 0x6DEA0000, %ebx = 0x9DEAFFFF, flag CF=1, ZF=0, OF=1, SF=1

The correct answer is: %eax = 0xD000FFFF, %ebx = 0x9DEA0000, flag CF=0, ZF=0, OF=1, SF=1

Question **81**

Not answered

Marked out of
5.00

Linking performed at compile time is known as

Answer:



The correct answer is: static linking

Question **82**

Not answered

Marked out of
5.00

Identify the command to invoke Linker on Linux system

Select one:

- ☐ 1. ddd
- ☐ 2. as
- ☐ 3. ld
- ☐ 4. gcc

The correct answer is: ld

Question 83

Not answered

Marked out of
20.00

Given a matrix $a[R][C]$ containing elements of type Double, where R is number of Rows and C is the number of Columns of the Matrix a.

You are required to access the element $a[x][y]$ of the matrix where x is the row index and y is the column index.

Assume that the matrix elements are stored in the main memory in Row-major order.

Write x86-64 assembly language instructions to get the memory address of the matrix element $a[x][y]$ in register %rax

Choose the correct program segment containing x86-64 assembly language instructions to compute the address of the location containing the matrix element $a[x][y]$.

Given Register Mapping: %rdi $\leftarrow$ a, %rax $\leftarrow$ C, %rcx $\leftarrow$ R, %rsi $\leftarrow$ y, %rbx $\leftarrow$ x

Select one:

- ☐ 1. leaq (%rax, 8), %rax
leaq (%rax, 8), %rax
mulq %rbx
leaq (%rbx, %rsi), %rax
- ☐ 2. leaq (%rdi, %rsi, 8), %rsi
mulq %rbx
leaq(%rax, %rsi, 8), %rax
- ☐ 3. leaq (%rdi, %rsi, 8), %rsi
mulq %rbx
leaq (%rsi, %rax, 8), %rax
- ☐ 4. leaq (%rax, 8), %rax
leaq(%rdi, %rsi, 8) %rsi
mulq %rax
leaq (%rax, %rsi), %rax

The correct answer is: leaq (%rdi, %rsi, 8), %rsi
mulq %rbx
leaq (%rsi, %rax, 8), %rax

Question 84

Not answered

Marked out of
5.00

Intel Corei7 processor chip introduced in 2008 has cores

Select one:

- ☐ 1. 4
- ☐ 2. 16
- ☐ 3. 8
- ☐ 4. 2

The correct answer is: 4

Question **85**
Not answered
Marked out of 5.00

..... is the first 32-bit intel processor referred to as IA32 introduced in 1985

Select one:

- ☐ 1. 80486
- ☐ 2. 80386
- ☐ 3. Pentium 4E
- ☐ 4. Pentium

The correct answer is: 80386

Question **86**
Not answered
Marked out of 5.00

..... is the first 16-bit intel processor introduced in 1978.

Select one:

- ☐ 1. 80186
- ☐ 2. 80286
- ☐ 3. 8086
- ☐ 4. 80386

The correct answer is: 8086

Question **87**
Not answered
Marked out of 20.00

Consider the following memory area containing matrix $a[4][5]$ elements stored in Column Major Order and write the address and value of the matrix element $a[2][3]$.

	Address	Data	Address	data
	-----	-----	-----	-----
a:	2000	c9	2010	9b
	2001	73	2011	df
	2002	49	2012	5e
	2003	ab	2013	47
	2004	ad	2014	83
	2005	fe	2015	c8
	2006	ed	2016	7e
	2007	ca	2017	5f
	2008	de	2018	7d
	2009	12	2019	9e

Select one:

- ☐ 1. address=2018, value=7d
- ☐ 2. address=2014, value=83
- ☐ 3. address=2006, value=ed
- ☐ 4. address=2010, value=9b

The correct answer is: address=2014, value=83

Question **88**
Not answered
Marked out of
5.00

..... is an explicit request made to the operating system to get a particular kernel service via interrupt.

Select one:

- ☐ 1. signal
- ☐ 2. system call
- ☐ 3. context switching
- ☐ 4. trap

The correct answer is: system call

Question **89**
Not answered
Marked out of
10.00

Identify intel 64-bit registers.....

Select one or more:

- ☐ 1. %rax
- ☐ 2. %eax
- ☐ 3. %ax
- ☐ 4. %rip

The correct answers are: %rax, %rip

Question **90**
Not answered
Marked out of
5.00

If %rbx = 0xa000 and %rsi = 8 then determine the Effective Address computed in the following instruction:

movq (%rbx, %rsi, 8), %rax

Select one:

- ☐ 1. 0xa00040
- ☐ 2. 0xa040
- ☐ 3. 0xa00064
- ☐ 4. 0xa064

The correct answer is: 0xa040

Question **91**
Not answered
Marked out of
20.00

Given the following structure, select the best ordering of the fields or elements of the structure to reduce memory space required for storing the structure.

```
struct S1 {  
    char c;  
    char d;  
    double v;  
    char e;  
    double w;  
    char f;  
} *p;
```

Select one:

- ☐ 1. struct S1 {
 char c;
 char d;
 double v;
 double w;
 char e;
 char f;
} *p;
- ☐ 2. struct S1 {
 char c;
 char d;
 double v;
 char e;
 char f;
 double w;
} *p;
- ☐ 3. struct S1 {
 char c;
 double v;
 char d;
 char e;
 double w;
 char f;
} *p;
- ☐ 4. struct S1 {

 double v;
 double w;
 char c;
 char d;
 char e;
 char f;
} *p;

The correct answer is: struct S1 {

```
    double v;  
    double w;  
    char c;  
    char d;  
    char e;  
    char f;  
} *p;
```

Question **92**

Not answered

Marked out of
10.00

Given a matrix $a[R][C]$ containing elements of type Double, where R is the number of Rows and C is the number of Columns of the Matrix a .

You are required to access the element $a[x][y]$ of the matrix where x is the row index and y is the column index.

Assume that the matrix elements are stored in the main memory in Row-major order.

Write the general expression to compute the address of the matrix element $a[x][y]$.

Select one:

- ☐ 1. $a + x * R * 8 + y * 8$
- ☐ 2. $a + i * C * 8 + j * 8$
- ☐ 3. $a + y * C * 8 + x * 8$
- ☐ 4. $a + x * C * 8 + y * 8$

The correct answer is: $a + x * C * 8 + y * 8$

Question **93**

Not answered

Marked out of
10.00

The `popq %rbx` instruction performs the following operations :

Select one:

- ☐ 1. `%rsp <-- %rsp + 8`
`%rbx <-- (%rsp)`
- ☐ 2. `%rbx <-- (%rsp)`
`%rsp <-- %rsp - 8`
- ☐ 3. `%rsp <-- %rsp - 8`
`%rbx <-- (%rsp)`
- ☐ 4. `%rbx <-- (%rsp)`
`%rsp <-- %rsp + 8`

The correct answer is: `%rbx <-- (%rsp)`
`%rsp <-- %rsp + 8`

Question **94**

Not answered

Marked out of
5.00

Given `%rdx = 0xFDEA0000`, `%rsi=100`, determine the value stored in register `%rax` after the execution of the following instruction :

`leaq (%rdx, %rsi), %rax`

Select one:

- ☐ 1. `0xFDEA0164`
- ☐ 2. `0xFDEA0064`
- ☐ 3. `0xFDEA0128`
- ☐ 4. `0xFDEA0100`

The correct answer is: `0xFDEA0064`

Question **95**

Not answered

Marked out of
10.00

Identify the commands to invoke Debugger on Linux system

Select one or more:

- ☐ 1. objdump
- ☐ 2. ddd
- ☐ 3. gdb
- ☐ 4. gcc

The correct answers are: gdb, ddd

Question **96**

Not answered

Marked out of
10.00

Consider the following memory area containing matrix a[4][5] elements stored in Row Major Order and write the address and value of the matrix element a[2][3].

	Address	Data	Address	data
	-----	-----	-----	-----
a:	2000	c9	2010	9b
	2001	73	2011	df
	2002	49	2012	5e
	2003	ab	2013	47
	2004	ad	2014	83
	2005	fe	2015	c8
	2006	ed	2016	7e
	2007	ca	2017	5f
	2008	de	2018	7d
	2009	12	2019	9e

Select one:

- ☐ 1. address=2014, value=83
- ☐ 2. address=2013, value=47
- ☐ 3. address=2009, value=12
- ☐ 4. address=2019, value=9e

The correct answer is: address=2013, value=47

Question **97**

Not answered

Marked out of
10.00

The pushq %rbx instruction performs the following operations :

Select one:

- ☐ 1. %rsp <-- %rsp - 8
(%rsp) <-- %rbx
- ☐ 2. (%rsp) <-- %rbx
%rsp <-- %rsp - 8
- ☐ 3. %rbx <-- (%rsp)
%rsp <-- %rsp - 8
- ☐ 4. %rsp <-- %rsp - 8
%rbx <-- (%rsp)

The correct answer is: %rsp <-- %rsp - 8
(%rsp) <-- %rbx

Question **98**

Not answered

Marked out of
10.00

Given %esi = 0x6DEAFFFF, %edi = 0x9DEA0000, determine the value stored in register %al after the execution of the following instructions :

```
cmpl %esi, %edi
setl %al
```

Select one:

- ☐ 1. 10000000
- ☐ 2. 00000000
- ☐ 3. 11111111
- ☐ 4. 00000001

The correct answer is: 00000001

Question **99**

Not answered

Marked out of
10.00

Write the function performed by the following assembly language program segment:

Registers: n in %rdx, result in %rax

```
movq $1, %rax
```

Loop:

```
imulq %rdx, %rax
```

```
subq $1, %rdx
```

```
cmpq $1, %rdx
```

```
jg Loop
```

```
ret
```

Select one:

- ☐ 1. square(n)
- ☐ 2. power(n)
- ☐ 3. Count the number of 1's in n
- ☐ 4. factorial(n)

The correct answer is: factorial(n)

Question **100**

Not answered

Marked out of
10.00

The condition checked in ja (jump on above) conditional jump instruction is

Select one:

- ☐ 1. ~CF&~ZF
- ☐ 2. ~(SF^OF)
- ☐ 3. (SF^OF)|ZF
- ☐ 4. ~(SF^OF)&~ZF

The correct answer is: ~CF&~ZF

Question **101**

Not answered

Marked out of
10.00

Given initially %rax=64 & %rbx=196, write the values in the registers %rax & %rbx after the execution of the following x86-64 assembly language program segment.

```
xorq  %rbx, %rax
xorq  %rax, %rbx
xorq  %rbx, %rax
```

Select one:

- ☐ 1. %rax= 64, %rbx = 196
- ☐ 2. %rax=64, %rbx=260
- ☐ 3. %rax = 196, %rbx = 64
- ☐ 4. %rax= 260, %rbx= 196

The correct answer is: %rax = 196, %rbx = 64

Question **102**

Not answered

Marked out of
5.00

Examples of Instruction Set Architecture

Select one:

- ☐ 1. Itanium
- ☐ 2. x86-64
- ☐ 3. ARM
- ☐ 4. All of the above

The correct answer is: All of the above

Question **103**

Not answered

Marked out of
5.00

A byte contains

Select one:

- ☐ 1. 32 bits
- ☐ 2. 64 bits
- ☐ 3. 8 bits
- ☐ 4. 16 bits

The correct answer is: 8 bits

Question **104**

Not answered

Marked out of
6.00

The condition checked in jge (jump on greater than or equal to) conditional jump instruction is

Select one:

- ☐ 1. $\sim(SF \wedge OF)$
- ☐ 2. $\sim(SF \wedge OF) \& \sim ZF$
- ☐ 3. $\sim(SF \wedge OF)$
- ☐ 4. $(SF \wedge OF) | ZF$

The correct answers are: $\sim(SF \wedge OF)$, $\sim(SF \wedge OF)$

Question **105**

Not answered

Marked out of
10.00

The instruction to perform add SIMD operation on Single-precision floating data is :

Select one:

- ☐ 1. addps %xmm0, %xmm1
- ☐ 2. addsd %xmm0, %xmm1
- ☐ 3. addpd %xmm0, %xmm1
- ☐ 4. addss %xmm0, %xmm1

The correct answer is: addps %xmm0, %xmm1

Question **106**

Not answered

Marked out of
10.00

Identify intel 32-bit registers

Select one or more:

- ☐ 1. %al
- ☐ 2. %ecx
- ☐ 3. %rbp
- ☐ 4. %esp

The correct answers are: %ecx, %esp

Question **107**

Not answered

Marked out of
10.00

Identify intel 16-bit registers

Select one or more:

- ☐ 1. %dl
- ☐ 2. %rsi
- ☐ 3. %di
- ☐ 4. %cx

The correct answers are: %cx, %di

Question **108**

Not answered

Marked out of
5.00

The YMM registers in AVX technology are all bits wide

Select one:

- ☐ 1. 256
- ☐ 2. 128
- ☐ 3. 512
- ☐ 4. 64

The correct answer is: 256

Question 109

Not answered

Marked out of
10.00

Consider the following memory area containing data and intel x86-64 assembly program segment given below:

Address	Data	Address	data
-----	-----	-----	-----
list: 2000	c9	2009	12
2001	73	2010	9b
2002	49	2011	df
2003	ab	2012	5e
2004	ad	2013	47
2005	fe	2014	83
2006	ed	2015	c8
2007	ca	2016	7e
2008	de	2017	5f

```
movq $list, %rbx
movq $2, %rsi
leaq (%rbx, %rsi, 4), %rax
movq (%rbx, %rsi, 4), %rcx
```

Determine the contents of the registers %rax and %rcx after executing the above assembly program segment.

Select one:

- ☐ 1. %rax=2008, %rcx= 0xde129bdf5e4783c8
- ☐ 2. %rax=2006, %rcx= 0xc883475edf9b12de
- ☐ 3. %rax=0xde129bdf5e4783c8, %rcx=2008
- ☐ 4. %rax=2008, %rcx= 0xc883475edf9b12de

The correct answer is: %rax=2008, %rcx= 0xc883475edf9b12de

Question **110**
Not answered
Marked out of
20.00

Write the function performed by the following assembly language program segment:

x in %rdi, %rax <-- result

```
movq $0, %rax
```

Loop:

```
movq %rdi, %rdx
```

```
andq $1, %rdx
```

```
addq %rdx, %rax
```

```
shrq %rdi
```

```
jne Loop
```

```
ret
```

Select one:

- ☐ 1. power(x)
- ☐ 2. factorial(x)
- ☐ 3. square(x)
- ☐ 4. Count the number of 1's in x

The correct answer is: Count the number of 1's in x

Question **111**
Not answered
Marked out of
10.00

What is the data stored in %rbx register after the execution of following instructions:

```
movq $0x8bcdef8960, %rbx
```

```
movb $0xbc, %bl
```

Select one:

- ☐ 1. 0x0000008bcdef89bc
- ☐ 2. 0xffffffffffffbc
- ☐ 3. 0x00000000000000bc
- ☐ 4. 0xfffff8bcdef89bc

The correct answer is: 0xfffff8bcdef89bc

Question **112**
Not answered
Marked out of
10.00

Write a single assembly language instruction which performs the same operation as performed by the following two instructions :

```
notq %rax
```

```
incq %rax
```

Answer: 

The correct answer is: negq %rax

Question **113**

Not answered
Marked out of
10.00

A is an array where entry is the address of a code segment implementing the action the program should take when the switch index equals i.

Answer: ✖

The correct answer is: jump table

Question **114**

Not answered
Marked out of
5.00

For gnu as assembler, assembler directive is used in assembly language program to reserve required number of bytes to store data.

Select one:

- ☐ 1. .single
- ☐ 2. .asciz
- ☐ 3. .quad
- ☐ 4. .skip

The correct answer is: .skip

Question **115**

Not answered
Marked out of
15.00

EM64T stands for

Answer: ✖

The correct answer is: Extended Memory 64-bit Technology

Question **116**

Not answered
Marked out of
10.00

Given the following assembly language program segment, write the value in register %rax after the execution of this program segment.

```
movq    $0x01289AB76EF34567, %rax
movb    $0xBB, %dl
movzbq   $dl, %rax
```

Select one:

- ☐ 1. %rax = 0x00000000000000BB
- ☐ 2. %rax = 0x01289AB76EF345BB
- ☐ 3. %rax = 0xFFFFFFFFFFFFBB
- ☐ 4. %rax = 0x01289AB76EF3BBBB

The correct answer is: %rax = 0x00000000000000BB

Question **117**

Not answered

Marked out of
10.00

Write the function performed by the following assembly language program segment:

`x in %rdx, y in %rbx, %rax <-- result`

`cmpq %rbx, %rdx`

`jle Else`

`subq %rbx, %rdx`

`movq %rdx, %rax`

`jmp Done`

Else:

`subq %rdx, %rbx`

`movq %rbx, %rax`

Done:

Select one:

- ☐ 1. $y - x$
- ☐ 2. $\text{mod}(x + y)$
- ☐ 3. $x - y$
- ☐ 4. $\text{abs}(x - y)$

The correct answer is: $\text{abs}(x - y)$

Question **118**

Not answered

Marked out of
5.00

What is the data stored in `%rbx` register after the execution of following instruction:

`movq $0x6def678a, %rbx`

Select one:

- ☐ 1. 0x000000006de678a9
- ☐ 2. 0x111111116de678a9
- ☐ 3. 0xffffffff6de678a9
- ☐ 4. 0x666666666de678a9

The correct answer is: 0x000000006de678a9

Question **119**

Not answered

Marked out of
5.00

File descriptor value associated with the keyboard (to be loaded into `%rdi`) is

Select one:

- ☐ 1. 60
- ☐ 2. 2
- ☐ 3. 0
- ☐ 4. 1

The correct answer is: 0

Question **120**

Not answered

Marked out of
10.00

Identify intel 8-bit registers

Select one or more:

- ☐ 1. %r8
- ☐ 2. %dh
- ☐ 3. %rbp
- ☐ 4. %bl

The correct answers are: %bl, %dh